

Group10-PA3 Formative Testing

Requirement 2 - Testing Plan, Results, Evaluation, and Points of Improvement

Field	Value
Group	Group 10
Project	FIFA.com Ticket Confidence + Chess.com Beginner Review
Testing method	Formative Testing
Reviewers	5 reviewers
Prototype alternatives	6
Selected directions	FIFA Alt 1 Status Dashboard Chess Alt 2 Card Review Mode
Prototype Demonstration Video	Covers all six paper prototypes https://www.youtube.com/watch?v=1DvikSQggyQ&feature=youtu.be

1. Testing Plan

1.1 Testing Objectives

ID	Objective	Signal captured
O1	Learnability	First action and entry-point discovery without instruction
O2	Task completion	Independent success, hesitation, failure, and stopping point
O3	Comprehension	Own-words explanation of ticket state or chess mistake
O4	Next-step clarity	Correct action, resolution, or practice step
O5	Trust / provenance	Freshness, official source, partner transition, and return path
O6	Cognitive load	Hesitation, repeated scanning, and confusion
O7	Learning continuity	Whether the interaction leads from explanation to practice
O8	Alternative differentiation	Distinct trade-offs across the interaction models
O9	Recovery	Ability to return, change direction, or continue after confusion
O10	Revision priority	Which changes are essential before PA4

1.2 Reviewer Profiles

We interviewed five reviewers who did not know anything about the website. Each reviewer worked from the same protocol and did not receive the other reviewers' results.

Reviewer	Novice profile	FIFA order	Chess order
R1	Higher digital literacy; low FIFA familiarity; no Chess analysis familiarity	Alt 1 -> Alt 2 -> Alt 3	Alt 2 -> Alt 1 -> Alt 3
R2	Careful reader; low FIFA and Chess.com familiarity	Alt 2 -> Alt 3 -> Alt 1	Alt 1 -> Alt 3 -> Alt 2
R3	Lower digital confidence; action-oriented; low domain familiarity	Alt 3 -> Alt 1 -> Alt 2	Alt 3 -> Alt 2 -> Alt 1
R4	Fast-scanning novice; low patience for long text	Alt 1 -> Alt 3 -> Alt 2	Alt 2 -> Alt 3 -> Alt 1
R5	Guided-workflow preference; moderate digital literacy	Alt 2 -> Alt 1 -> Alt 3	Alt 1 -> Alt 2 -> Alt 3

1.3 Testing Method

We interviewed five reviewers who did not have any knowledge about the website.

Each reviewer evaluated all six paper prototypes using the same scenarios and task definitions. Prototype order varied between reviewers to reduce order effects.

For each task, we recorded Independent Success, Success with Hesitation, or Failure, together with observations, feedback, and suggested improvements.

1.4 Test Order and Bias Control

The five orders rotate the alternatives across both scenarios. Each alternative was reset to the same paper context before review, and the protocol used the same task wording, neutral probes, success criteria, issue categories, and outcome coding for every reviewer. Preference was recorded after the task sequence as a supporting signal rather than a standalone winner rule.

1.5 FIFA Scenario and Tasks

Scenario: You have tickets connected to two football events. One event is confirmed and one is pending. Understand whether everything is okay, what happens next, and how to perform a common ticket action before travelling.

ID	Task	Success signal
F1	Find the confirmed event	Correctly identifies the confirmed event or order
F2	Find the pending event	Correctly identifies what still requires attention
F3	Explain the current state	Explains the state in own words
F4	Identify what happens next	Finds the next expected step or timing
F5	Check information freshness	Finds last-updated or current-information evidence
F6	Judge official-source trust	Identifies the official source signal
F7	Find ticket details	Finds the relevant ticket-detail action
F8	Transfer a ticket	Finds and interprets the transfer action
F9	Add the event to a calendar	Finds the calendar action and understands its scope
F10	Recover from an unresolved state	Returns to a useful status path and identifies remaining support or resolution

1.6 FIFA Functionalities Under Test

Alternative	Functionality
Alt 1 - Status Dashboard	Status cards, confirmed/pending distinction, event list, primary next action, quick actions, support, current-state summary
Alt 2 - Timeline Tracker	Completed/current/upcoming stages, progress message, freshness and official-source cues, update history
Alt 3 - Action Hub	Quick actions, ticket-management shortcuts, official options, support/security, partner transition cue

1.7 Chess Scenario and Tasks

Scenario: You have finished a chess game. You know that you made mistakes, but you are not familiar with analysis tools. Understand one important mistake, learn a better move, and decide what to practise next.

ID	Task	Success signal
C1	Find where to begin	Starts review without analysis-tool instruction
C2	Identify one mistake	Selects an important moment or explanation
C3	Explain why	Describes why the move was a mistake in own words
C4	Find the better move	Locates and interprets the suggested move
C5	Try the better move	Uses the move-trial interaction
C6	Find practice	Finds a relevant practice bridge
C7	Ask for or inspect more explanation	Uses depth or follow-up support appropriately
C8	Choose another review moment	Uses control to continue or change the review
C9	Return to a useful next step	Identifies what to do next after exploring

1.8 Chess Functionalities Under Test

Alternative	Functionality
Alt 1 - Beginner Review Flow	Guided sequence, progress indicator, plain-language explanation, better move, practice bridge, optional depth
Alt 2 - Card Review Mode	Review-moment cards, mini-board scan, selected-card explanation, user-controlled priority, practice entry
Alt 3 - Side-by-Side AI Assistant	Question entry, assistant explanation, board highlights, context awareness, follow-up prompts, practice action

1.9 Success Criteria

Criterion	Pass signal	Revise if
Independent completion	At least 4 of 5 complete a core task without rescue.	Fewer than 4 complete or the same blocker stops 2+ reviewers.
Own-words comprehension	At least 4 of 5 explain the state or mistake.	Two or more form the same critical wrong mental model.
Next-step clarity	At least 4 of 5 identify the intended action or practice step.	Ownership, destination, or relevance stays ambiguous.
Intervention	No more than 1 of 5 needs intervention on a core task.	Two or more interventions occur at the same step.
Trust / practice continuity	Freshness/source/partner transition or practice bridge is understood.	Source, return path, or practice action is invisible.

1.10 Data Collection

Capture	Record	Why it matters
Reviewer + order	Reviewer ID, profile, scenario, alternative, and order position	Separates design effect from learning effect
Task result	Independent Success / Hesitation / Failure and stopping point	Shows effectiveness and rescue burden
Observation	First action, hesitation, wrong path, and recovery	Locates cognitive-load hotspots
Feedback	Clear, confusing, missing, or unnecessary; clarity/confidence rating	Captures perception without replacing behavior
Issue + revision	Category, severity, problem -> cause -> recommendation	Turns evidence into PA4 change candidates
Comparison	Preference, reason, strength, weakness, and unresolved concern	Supports synthesis; not a standalone selection rule

1.11 Issue Severity

Issue categories include entry and learnability, comprehension, hierarchy, terminology, navigation, ownership, trust/provenance, action visibility, practice continuity, control/flexibility, and recovery. Critical breaks the core task or trust boundary; High is a repeated blocker or wrong mental model; Medium makes the task possible but inefficient or unclear; Low is polish or optional detail.

2. Completed Testing Results

The 19 task definitions were applied across the six alternatives.

FIFA contributes 150 outcomes and Chess contributes 135 outcomes.

2.1 R1 summary

Scenario	Alt 1	Alt 2	Alt 3	Preference
FIFA	7 / 2 / 1	5 / 2 / 3	8 / 0 / 2	Alt 1
Chess	8 / 1 / 0	7 / 2 / 0	6 / 2 / 1	Alt 2

Outcome cells show Independent / Hesitation / Failure.

2.2 R2 summary

Scenario	Alt 1	Alt 2	Alt 3	Preference
FIFA	7 / 2 / 1	5 / 2 / 3	8 / 0 / 2	Alt 2
Chess	7 / 2 / 0	7 / 2 / 0	4 / 4 / 1	Alt 2

Outcome cells show Independent / Hesitation / Failure.

2.3 R3 summary

Scenario	Alt 1	Alt 2	Alt 3	Preference
FIFA	5 / 3 / 2	5 / 2 / 3	7 / 1 / 2	Alt 1
Chess	7 / 2 / 0	7 / 2 / 0	5 / 3 / 1	Alt 2

Outcome cells show Independent / Hesitation / Failure.

2.4 R4 summary

Scenario	Alt 1	Alt 2	Alt 3	Preference
FIFA	6 / 2 / 2	6 / 1 / 3	7 / 1 / 2	Alt 1
Chess	7 / 2 / 0	7 / 2 / 0	6 / 2 / 1	Alt 2

Outcome cells show Independent / Hesitation / Failure.

2.5 R5 summary

Scenario	Alt 1	Alt 2	Alt 3	Preference
FIFA	6 / 2 / 2	5 / 2 / 3	7 / 1 / 2	Alt 2
Chess	7 / 1 / 1	6 / 3 / 0	5 / 3 / 1	Alt 2

Outcome cells show Independent / Hesitation / Failure.

2.6 Cross-reviewer aggregate

Scenario	Alternative	Independent	Hesitation	Failure	Total
FIFA	Alt 1 - Status Dashboard	31	11	8	50
FIFA	Alt 2 - Timeline Tracker	26	9	15	50
FIFA	Alt 3 - Action Hub	37	3	10	50
Chess	Alt 1 - Beginner Review Flow	34	11	0	45
Chess	Alt 2 - Card Review Mode	36	8	1	45
Chess	Alt 3 - Side-by-Side AI Assistant	26	14	5	45

Scenario	Preference signal
FIFA	R1 Alt 1; R2 Alt 2; R3 Alt 1; R4 Alt 1; R5 Alt 2 -> Alt 1 3/5, Alt 2 2/5, Alt 3 0/5
Chess	R1-R5: Alt 2 -> Alt 1 0/5, Alt 2 5/5, Alt 3 0/5

3. Evaluation and Selection

3.1 FIFA Alternative Comparison

Alternative	I / H / F	Evidence interpretation	Selection role
Alt 1 - Status Dashboard	31 / 11 / 8	Compact status-first scan; recurring Pending reason, owner, timing, and source/partner-transition gaps; 3/5 preference.	Selected; revise without changing the stable entry.
Alt 2 - Timeline Tracker	26 / 9 / 15	Strong progress and freshness cues, but ticket actions and ownership remain unclear; 2/5 preference.	Borrow timeline and ownership cues.
Alt 3 - Action Hub	37 / 3 / 10	Fastest independent completion, but status context and the partner boundary can be skipped; 0/5 preference.	Borrow action grouping and partner-transition treatment.

3.2 Selected FIFA Direction

FIFA Alt 1, Status Dashboard, was selected.

Alt 3 produced more independent task successes, but Alt 1 gave a better overall balance of status visibility, next-step clarity, action access, and reviewer preference. Its main weaknesses also had focused fixes, such as explaining Pending and clarifying the next action.

3.3 Chess Alternative Comparison

Alternative	I / H / F	Evidence interpretation	Selection role
Alt 1 - Beginner Review Flow	34 / 11 / 0	Best entry, explanation-to-practice continuity, and 0/5 preference; remaining risk is recovery and depth visibility.	Selected; revise without changing the guided sequence.
Alt 2 - Card Review Mode	36 / 8 / 1	Low failure and strong scan/control, but the first learning priority is ambiguous; 5/5 preference.	Borrow a Start here and priority cue.
Alt 3 - Side-by-Side AI Assistant	26 / 14 / 5	Flexible prompts help, but blank input and continuation/scope create friction; 0/5 preference.	Borrow bounded prompts and follow-up help.

3.4 Selected Chess Direction

Chess Alt 2, Card Review Mode, was selected. It produced 36 independent successes, only one failure, and was preferred by all five reviewers. Its guided mistake-to-practice sequence matched the beginner scenario more closely than the other alternatives.

3.5 Selection Method

Selection combines task outcomes, recurring issue severity and recurrence, comprehension and next-step clarity, trust/provenance or practice continuity, error risk, preference, and revision effort. Preference alone does not select a direction.

4. Points of Improvement

The recommendations below keep the strongest interaction model in each selected direction and address recurring problems identified by the reviewers.

4.1 FIFA selected-prototype revisions

ID	Evidence-linked change	Trace	Recommended change
F-01	Pending card gains a reason.	R1-R5 recurring Pending ambiguity; especially F4/F10 observations and feedback.	Add 'Awaiting FIFA confirmation' beside Pending.
F-02	Pending card gains an owner.	R1-R5 recurring ownership gap in pending-state observations.	State 'Owner: FIFA' so system progress is not read as user work.
F-03	Pending card gains expected timing.	R1-R5 F4 next-step and timing friction.	Add 'Expected timing: next confirmation update.'
F-04	Show whether the user needs to act.	R1-R5 F2/F4/F10 confusion about whether the reviewer must act.	Add 'No action is required from you right now' to the Pending state.
F-05	Add the official FIFA source beside freshness.	R1/R3/R4/R5 trust and provenance observations around F5/F6.	Place the source label beside 'Last updated'.
F-06	Clarify View Order versus View Tickets.	R1/R2/R3/R5 action-label ambiguity around F7.	Keep both actions but state their different destinations.
F-07	Add a per-action partner cue.	R1-R5 F10 partner-transition and external-service warning issues.	Show the partner destination before Continue.
F-08	Keep the compact status-first overview.	R1/R3/R4/R5 status-scan strength and preference evidence.	Keep the summary, event card, and action grouping as the main entry.

4.2 Chess selected-prototype revisions

ID	Evidence-linked change	Trace	Recommended change
C-01	Make Start Beginner Review explicit.	R1-R5 C1 entry observations; Alt 1 was preferred by all five reviewers.	Make the first action a visible Start Beginner Review control.
C-02	Separate review-step and game-move labels.	R1-R5 C1/C2 entry and orientation notes.	Show 'Review step 1 of 3 - Game move 23'.
C-03	Pair notation with plain-language movement.	R1-R5 C3 vocabulary and explanation feedback.	Pair Qe2 with 'move the queen to e2 to protect the pawn'.
C-04	Add another explanation and help controls.	R1-R5 C7/C9 recovery and depth-support observations.	Keep explicit support and alternate-explanation controls.
C-05	Show feedback after trying the move.	R1-R5 C5/C6 feedback and practice-continuity observations.	State the changed board result after Qe2 is tried.
C-06	Keep Practice this idea visible.	R1-R5 C6 practice-bridge issues and Alt 1 strength.	Keep the practice action next to the explanation and result.
C-07	Keep Next mistake and recovery visible.	R1-R5 C8/C9 continuation and recovery observations.	Keep next-mistake and recovery actions visible after exploration.
C-08	Keep the one-mistake-at-a-time sequence.	R1-R5 overall comparison; Alt 1 had the clearest learning continuity.	Keep the guided sequence as the selected core path.

4.3 Improvement principles

- Keep the selected prototype's strongest interaction model and revise the recurring blocker rather than replacing the design hypothesis.
- Make ownership, timing, provenance, and partner transition visible where a reviewer previously had to infer them.
- Make the novice learning path explicit: entry, cause, move, result, practice, help, and recovery.

- Borrow a cue from a non-selected alternative only when it solves a traced issue without importing its larger trade-off.

6. Conclusion

Five reviewers evaluated six prototypes using 19 task definitions.

FIFA Alt 1 Status Dashboard and **Chess Alt 2** Card Review Mode were selected, and the evaluation identified focused points of improvement for both directions.